

GenerateTemplate

`GenerateTemplate` [*directory*, *name*, *a*, *θ* , *imageSize*,
maxBardeenCoordinate, *shellRadius*_: 50, *radiusLimit*_: 0, *Options*]

generates a template containing information about null geodesics specified by (dimensionless) *a*, *θ* (distant observer's θ) and maximal Bardeen coordinate that is shown. The number of geodesics to be generated is specified in *imageSize* in the form {xsize, ysize}. The template is saved in *directory/name.mx* file, which can be used later with `DiskImageFromTemplate`.

`GenerateTemplate`[*directory_*, *name_*, *a_*, *$\theta_$* , *imageSize_*, *maxBardeenCoordinate_*, *shellRadius*_:50, *radiusLimit*_:0, *OptionsPattern*[]] is a function, which takes the name of the directory, in which a template should be saved, *directory*, a name of the template's file *name*, the dimensionless angular momentum of the black hole *a*, the observer's polar angle *θ* , a list of length two containing the size of the image to be generated from the template in pixels *imageSize*, the maximal Bardeen coordinate *maxBardeenCoordinate*, the optional arguments *shellRadius* (in $G=c=M=1$ units) which dictates the radius of shell intersection coordinates which are used for generating distorted stellar background using the `StellarBackgroundFromTemplate` function, *radiusLimit*, the greatest radius at which the disk near the black hole should be visible, and an options pattern.

The following options can be given:

"Rotation"	"Counterclockwise"	Sets the direction of rotation of the black hole. The default option is "Rotation"→"Counterclockwise". The opposite is "Rotation"→"Clockwise".
"PhiRange"	{-Pi, Pi}	Sets the range of output of the azimuthal angle. The default is "PhiRange"→{-∞, ∞}, which starts the coordinate at 0 and does not take the modulus of it after full windings. Typical options could be {-π, π} or {0, 2π}, but other option values in the format {bottomvalue, topvalue} are valid as well.

Tech Notes ⓘ
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Related Links ⓘ
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See Also ⓘ
KerrNullGeoDistant ▪ **DiskImageFromTemplate** ▪ **DiskImage** ▪
StellarBackgroundFromTemplate ▪ ⓘ

Related Guides
KerrImages

Examples Initialization ⓘ

Needs ["BlackHoleImages`"]

Basic Examples

[More Examples ▸](#)

Generate a template of a geometry given by the spin parameter $a=0.5$ with the observer at $\theta_0=0.45\pi$. Generate 100 x 100 points with maximal Bardeen coordinate 20:

```
In[7]:= GenerateTemplate[Directory[], "template_a0.5_th0.45pi_size100x100_mBC20", 0.5, 0.45  $\pi$ , {100, 100}, 20]
```

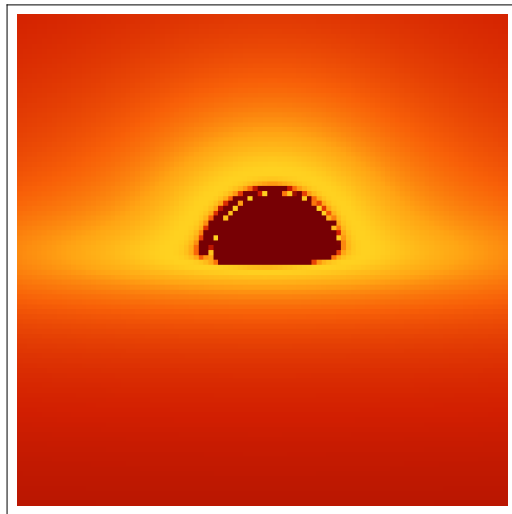
```
%
```

From the generated template, generate an image of the disk's physical temperature using the `DiskImageFromTemplate` function. The black hole has a solar mass, the matter influx is 10^{14} kg/s and $\alpha=0.1$:

```
In[10]:= img = DiskImageFromTemplate[Directory[] <> "\\template_a0.5_th0.45pi_size100x100_mBC20.mx",  
0.5, 0.1, 1500, 1, "Grid" -> False, "Output" -> {"PhysicalTemperature"}]
```

```
In[11]:= ArrayPlot[Reverse[img[[1]]], ColorFunction -> "SolarColors"]
```

Out[11]=



More Examples ⓘ[Scope](#)[Generalizations & Extensions](#)[Options](#)["Rotation"](#)["PhiRange"](#)[Applications](#)[Properties & Relations](#)[Possible Issues](#)[Interactive Examples](#)[Neat Examples](#)

Metadata

New in: XX | Modified in: | Obsolete in:

Categorization ⓘ

Keywords

Syntax Templates